# E-Commerce Sales Analysis

## 1. Problem Statement

The objective of this task is to analyze the e-commerce sales dataset and identify overall business performance, high-performing and loss-making areas, regional sales patterns, and the impact of discounting on profitability. The analysis aims to convert raw transactional data into actionable business insights through exploratory data analysis and dashboarding.

## 2. Objective

The goal of this analysis is to understand how the business is performing overall, identify which categories, sub-categories, segments, and regions are performing well or poorly, and provide practical recommendations based on the findings.

## 3. Dataset Overview & Data Cleaning

- The original dataset had 9,994 rows and 13 columns.
- Columns included: Ship Mode, Segment, Country, City, State, Postal Code, Region, Category, Sub-Category, Sales, Quantity, Discount, and Profit.
- There were no missing (null) values in any column, so no rows had to be dropped for that reason.
- I found 17 duplicate rows (rows that were exactly repeated) and removed them. After this, the dataset had 9,977 rows.
- The Postal Code column was originally read as a number, but I changed it to Text, since postal codes are labels/IDs and should not be added or averaged.
- Note: The dataset does not have an Order Date column or a Product Name column. Because of this, I could not do a month-wise trend analysis or a true product-level analysis. I adapted the analysis at the Category/Sub-Category level using the available fields.

## 4. Approach / Methodology

The analysis was performed in the following stages:

1. Data understanding and data cleaning
2. Duplicate check and data type validation
3. KPI calculation (Sales, Profit, Quantity, and Profit Margin)
4. Category, sub-category, region, and segment analysis
5. Discount vs Profit analysis
6. Dashboard creation in Power BI
7. Insight extraction and business recommendations

## 5. Key Performance Indicators (KPIs)

| KPI | Value |
|---|---|
| Total Sales | $2,296,195.59 |
| Total Profit | $286,241.42 |
| Total Quantity Sold | 37,820 units |
| Overall Profit Margin | 12.47% |

A profit margin of 12.47% means that for every $100 in sales, the business is keeping around $12.5 as profit. This is a decent overall margin, but it varies significantly across categories, sub-categories, and regions.

# 6. Category-Wise Performance

| Category | Sales ($) | Profit ($) | Profit Margin |
|---|---|---|---|
| Technology | 836,154.03 | 145,454.95 | 17.4% |
| Office Supplies | 718,735.24 | 122,364.66 | 17.0% |
| Furniture | 741,306.31 | 18,421.81 | 2.5% |

Furniture has almost the same sales as Office Supplies, but its profit is much lower (only 2.5% margin compared to 17% for the other two). This indicates that Furniture is selling well but is not generating comparable profit, most likely due to heavy discounting and weaker margin structure.

# 7. Sub-Category Analysis

## 7.1 Sub-Categories Making a Loss

| Sub-Category | Sales ($) | Profit ($) |
|---|---|---|
| Tables | 206,965.53 | -17,725.48 |
| Bookcases | 114,880.00 | -3,472.56 |
| Supplies | 46,673.54 | -1,189.10 |

Tables is the biggest problem area. It records a loss of about $17,725 even though it is one of the higher-selling sub-categories. This means Tables is being sold in large volume but is still reducing overall profitability.

## 7.2 Top 10 Sub-Categories by Sales

| Rank | Sub-Category | Sales ($) |
|---|---|---|
| 1 | Phones | 330,007.05 |
| 2 | Chairs | 327,777.76 |
| 3 | Storage | 223,843.61 |
| 4 | Tables | 206,965.53 |
| 5 | Binders | 203,409.17 |
| 6 | Machines | 189,238.63 |
| 7 | Accessories | 167,380.32 |
| 8 | Copiers | 149,528.03 |
| 9 | Bookcases | 114,880.00 |
| 10 | Appliances | 107,532.16 |

Phones and Chairs bring in the most sales. It is notable that Tables and Bookcases appear in this top 10 list but are also in the loss-making list above, showing that high sales do not always translate into high profit.

# 8. Region-Wise Performance

| Region | Sales ($) | Profit ($) |
|---|---|---|
| West | 725,255.64 | 108,329.81 |
| East | 678,435.20 | 91,506.31 |
| South | 391,721.90 | 46,749.43 |
| Central | 500,782.85 | 39,655.88 |

West is the best-performing region in terms of both sales and profit. Central is the weakest from a profit perspective — even though Central has more sales than South, its profit is lower than South's. This suggests that discounting or cost/margin structure in Central may be affecting profitability.

# 9. Segment-Wise Performance

| Segment | Sales ($) | Profit ($) |
| --- | --- | --- |
| Consumer | 1,160,832.77 | 134,007.44 |
| Corporate | 706,070.13 | 91,954.98 |
| Home Office | 429,292.68 | 60,279.00 |

The Consumer segment brings in the highest sales and profit, followed by Corporate and then Home Office.

## 10. Effect of Discount on Profit

I checked whether there is a relationship between Discount and Profit. The correlation came out to be approximately -0.22, which indicates that as discount increases, profit tends to decrease.

To understand this further, I compared the average discount given on profitable orders vs loss-making orders:

| Order Type | Average Discount |
| --- | --- |
| Profitable orders (Profit ≥ 0) | 8.1% |
| Loss-making orders (Profit < 0) | 48.1% |

This is a major difference — loss-making orders have almost six times more discount on average than profitable orders. This strongly suggests that high discounts are a major reason behind the losses, especially for Tables, Bookcases, and Furniture in general.

## 11. Key Findings

- The business is profitable overall, with a profit margin of 12.47%, but profitability is not evenly distributed across all categories.
- Furniture has good sales but a very low margin (2.5%), mainly because of Tables and Bookcases.
- Tables is the single biggest loss-maker, even though it sells quite well.
- West is the best-performing region, while Central is the weakest in terms of profit despite having higher sales than South.
- Orders with heavy discounts (48% on average) are much more likely to result in a loss compared to orders with low discounts (8% on average).

## 12. Visualizations

A Power BI dashboard was created to present the key findings of the analysis. The dashboard includes KPI cards for Total Sales, Total Profit, Total Quantity Sold, and Profit Margin, along with visuals for Sales by Region, Profit by Category, Top Sub-Categories by Sales, and Profit by Sub-Category. The dashboard also includes slicers for Category, Region, and Segment to allow interactive filtering.

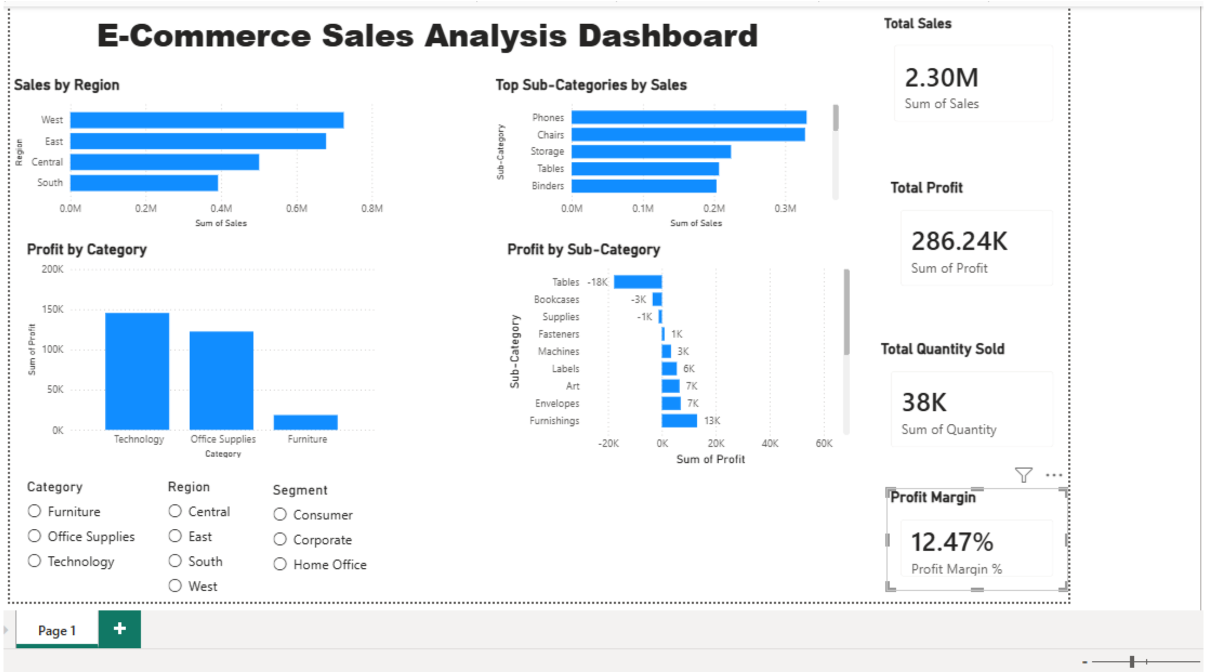


*Figure 1: E-Commerce Sales Analysis Dashboard (Power BI)*

## 13. Recommendations

- Reduce or control the discounts given on Tables and Bookcases, since these appear to be the most affected by over-discounting.
- Review why Furniture has such a low margin overall — likely reasons could include higher discounting, pricing issues, or product-level margin structure.
- Investigate why the Central region has lower profit than South despite having higher sales. This may be linked to discounting patterns or weaker margin realization.
- Continue investing in Phones, Chairs, and Copiers, as these sub-categories already show strong sales and healthy profit contribution.
- For future analysis, it would help to have an Order Date column (for trend analysis) and a Product Name column (for true product-level insights).

## 14. Conclusion

Overall, the business is performing well and is profitable, but profit is not spread evenly across all categories and sub-categories. Furniture — and Tables in particular — is where a large share of the losses is coming from, and this appears to be linked more to high discounting than to low demand.

If discounting on these specific sub-categories is controlled and weaker-profit regions are reviewed more closely, the overall profit margin could improve without necessarily increasing total sales.